

US IT Recruitment Analytics Project

1. Business Problem

TechStaff Solutions lacks a centralized analytical view of its US IT recruitment operations. Management needs to understand recruitment demand, candidate application patterns, sourcing-channel performance, offer acceptance, compensation, and recruitment process efficiency.

The organization currently has recruitment data across candidates, jobs, applications, interviews, offers, and companies. However, without structured analysis and visualization, it is difficult for management to identify recruitment bottlenecks, understand hiring demand, evaluate sourcing channels, and make data-driven decisions.

The goal of this project is to analyze the recruitment data using SQL, Python, Excel, and Power BI to identify actionable insights and recommend improvements to the recruitment process.

2. Business Objectives

The primary objectives of this project are:

1. **Analyze recruitment demand** by identifying the most active job roles, role families, job levels, and work arrangements.
2. **Evaluate candidate application patterns** to understand candidate interest and competition across different recruitment categories.
3. **Analyze recruitment source performance** to identify which sourcing channels generate stronger candidate and offer outcomes.
4. **Evaluate offer performance** by analyzing offer volume, offer acceptance rate, and acceptance patterns across different sources and job levels.
5. **Analyze compensation trends** by comparing average offer salaries across different job levels.
6. **Measure recruitment process efficiency** by analyzing the time taken from application to offer.
7. **Identify recruitment bottlenecks and opportunities** and provide data-driven recommendations to improve recruitment effectiveness.

3. Stakeholders

The key stakeholders who would use the insights from this recruitment analytics project are:

Stakeholder	Information Needed
Recruitment Manager	Overall recruitment performance, funnel efficiency, offer acceptance, and sourcing performance
Recruiting Team Lead	Candidate pipeline, sourcing channels, role demand, and recruitment bottlenecks
Recruiters	Candidate demand, job requirements, sourcing effectiveness, and offer outcomes
Operations Manager	Recruitment process efficiency, workload, and operational performance
Business Development / Account Managers	Client hiring demand, job requirements, and recruitment performance by client
Senior Management	Overall hiring trends, compensation, recruitment efficiency, and strategic recommendations

4. Business Questions

The analysis aims to answer the following key business questions:

1. Which **job role families** have the highest hiring demand?
2. Which **job levels** receive the most candidate applications?
3. Which **recruitment sources** generate the highest application volume?
4. Which recruitment sources have the **highest offer acceptance rates**?
5. What is the **average offer salary** across different job levels?
6. Which **work arrangements** (remote, hybrid, onsite) attract the most applications?
7. How does **candidate experience** vary across the candidate pool?
8. How many applications are generated per available job across different role families?
9. What is the **average time from application to offer**?
10. What major **recruitment bottlenecks and opportunities** can be identified from the data?

5. KPIs & Metrics

The following KPIs were selected to measure recruitment volume, conversion, compensation, and process efficiency:

KPI	Definition	Value
Total Candidates	Unique candidates in the recruitment database	9,000
Total Jobs	Total job postings/requisitions	2,000
Total Applications	Total candidate applications	15,000
Total Interviews	Total recorded interviews	7,196
Total Offers	Total offers generated	938
Accepted Offers	Offers accepted by candidates	699
Offer Acceptance Rate	Accepted offers ÷ total offers	74.52%
Average Offer Salary	Average salary offered to candidates	\$129,879.69
Application-to-Offer Time	Average days between application and offer	28.70 days
Applications per Job	Total applications ÷ total jobs	7.50

6. Dataset & Data Sources

Dataset

This project uses a synthetic **US IT Job Postings & Applications** dataset containing recruitment-related information across candidates, companies, job postings, applications, interviews, and offers.

The dataset was selected because its structure closely represents a real-world recruitment analytics environment and allows analysis of the recruitment lifecycle from job demand through offer acceptance.

Dataset Size

Table	Records
Companies	320
Candidates	9,000
Jobs	2,000
Applications	15,000
Interviews	7,196

Table	Records
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Offers	938
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Main Data Entities

- **Companies** — company information and industry
- **Candidates** — candidate experience, location, and skill information
- **Jobs** — job postings, role family, level, salary, location type, and employment type
- **Applications** — candidate applications, source, stage, and score
- **Interviews** — interview rounds and outcomes
- **Offers** — offered salary, acceptance status, and offer dates

Data Source

The dataset was obtained from the **Hugging Face** platform and was created as a synthetic, privacy-safe dataset for analytics and machine learning use cases.

Tools Used

- **Excel** — data inspection, cleaning, and validation
- **Python** — data cleaning, exploratory data analysis, and validation
- **MySQL** — relational data modeling and SQL analysis
- **Power BI** — data modeling, KPI development, visualization, and dashboarding

- **7. Data Quality Audit**

- Before performing analysis, the dataset was audited to identify potential data quality issues affecting the reliability of the analysis.
- The audit covered the following areas:

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|--------------------------------|---------------------------------------|
| • Data Quality Check | • Result |
| • Duplicate primary IDs | • No duplicates found |
| • Missing values | • Reviewed and handled where required |
| • Data types | • Checked and standardized |

- **Data Quality Check**
- **Result**
- **Referential integrity**
 - Relationships between related tables validated
- **Invalid salary values**
 - Checked; offer salaries were positive
- **Offer acceptance dates**
 - Validated against acceptance status
- **Date consistency**
 - Date fields reviewed and standardized
- **Categorical values**
 - Reviewed for consistency
- **Business-rule validation**
 - Key recruitment rules checked
- **Key Data Quality Findings**
- Primary identifiers across the six tables contained **no duplicate records**.
- Relationships between candidates, jobs, applications, interviews, and offers were validated.
- accepted_at contains null values for offers that were not accepted. These were **not artificially filled**.
- An acceptance_status field was created to clearly distinguish **Accepted** and **Not Accepted** offers.
- Boolean acceptance values were standardized for database analysis.
- Date fields were standardized to support monthly and time-based analysis.
- **Data Quality Outcome**
- The audit confirmed that the dataset was suitable for further **data cleaning, SQL analysis, Python EDA, and Power BI reporting**, subject to the documented limitations.